Suppose a force acts on an object which is moving in a line. The work is the energy transfered to the object by the force. If the force is constant, then the work equals

force · displacement

Suppose a force acts on an object which is moving in a line. The **work** is the energy transfered to the object by the force. If the force is **constant**, then the work equals

$$\text{force} \cdot \text{displacement}$$

Suppose the force is not constant, and is instead a function $F(x)$ of the position x. Then the work equals

$$\int_a^b F(x)\,dx$$

where a and b are start and end values of x.

Example: I hold a 2 lb ball 3 ft above the ground. I throw the ball upward with a constant 7 lb of force, and it leaves my hand at 5 ft above the ground.

Work done by me on ball = (7 lb)(2 ft)

= 14 ft·lb

Example: I hold a 2 lb ball 3 ft above the ground. I throw the ball upward ~~with a constant 7 lb of force~~, and it leaves my hand at 5 ft above the ground.

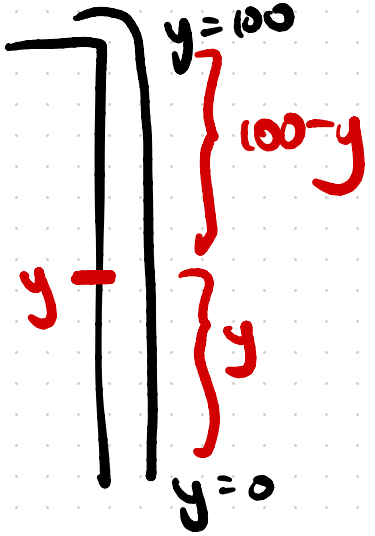
Suppose instead the force is given by the equation $F(y) = y + 2$, where y is the height above the ground.

Work done by me on ball

$$\int_3^5 F(y)\,dy = \int_3^5 (y+2)\,dy = \frac{1}{2}y^2 + 2y \Big|_3^5$$

$$= \frac{25}{2} + 10 - \left(\frac{9}{2} + 6\right) = 12$$

A 100 ft long, 200 lb rope hangs from a cliff. How much work is needed to pull the rope halfway up the cliff?



Subdivide rope into pieces with "length" dy.

Weight density of rope is 2 lb/ft

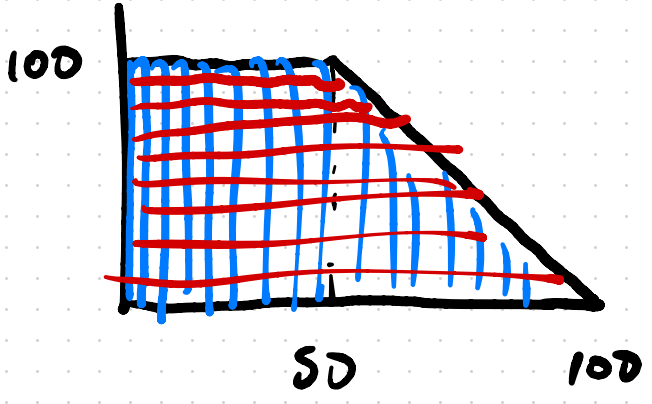
Weight of one piece is $2\,dy$

The piece at height y needs to move

$$\begin{cases} 50 & \text{if } y \le 50 \\ 100-y & \text{if } y \ge 50 \end{cases}$$

Total work

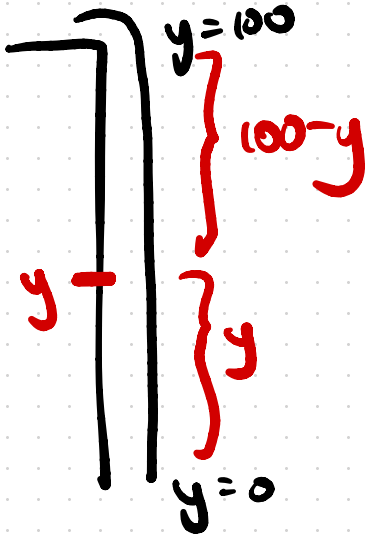
$$\int_0^{50} 50(2dy) + \int_{50}^{100} (100-y)(2dy)$$



$$50 \cdot 100 + \frac{1}{2} 50 \cdot 100$$

$$7500$$

A 100 ft long, 200 lb rope hangs from a cliff. How much work is needed to pull the rope halfway up the cliff?



Treat the rope as a particle starting at $y=0$, with the entire weight concentrated on the particle.

As the rope/particle moves, the weight of the particle changes to equal the actual weight of the rope.

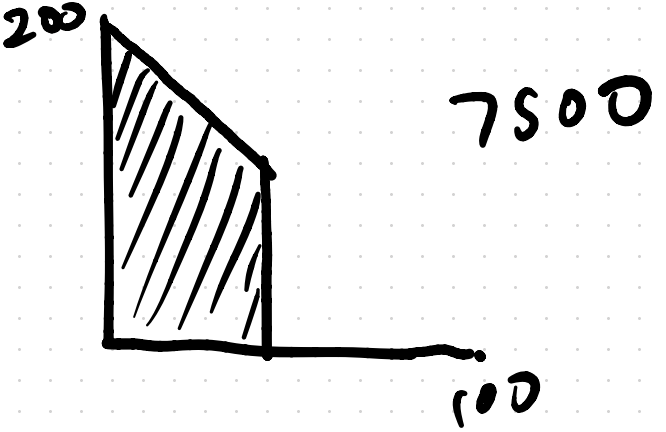
Weight of the rope/particle at height y $= 2(100-y)$

length of rope

Work done on particle y

$$\int_0^{50} 2(100-y)\,dy$$

force at height y



7500

5. (12 points) A 50 ft rope weighing 2 lb/ft is lifting a 5 lb bucket intially carrying 8 lb of water. It is carrying the water up a well that is also 50 ft. The bucket of water is being lifted up at a rate of 3 ft/sec. The bucket has a hole on the bottom, causing water to leak out at a rate of 1/4 lb/sec. Find the work required to lift the rope and leaky bucket of water up to the top of the well.

[Hint: find the total weight $F(y)$ in pounds of the rope and leaky bucket as a function of distance y from the bottom of the well.]



$F(y)$ = weight of rope + weight of bucket + weight of water

weight of rope $= 2(50-y)$

weight of bucket $= 5$

weight of water $= 8 - \frac{y}{12}$

How much water has leaked when the bucket is at height y?

How much time has elapsed when the bucket is at height y?

distance = rate · time

y ft $= (3$ ft/s$)\, t$

$\frac{y}{3}$ s $= t$

Leaked water = $(\frac{1}{4}$ lb/s$)(\frac{y}{3}$ s$)$

$= \frac{y}{12}$ lb

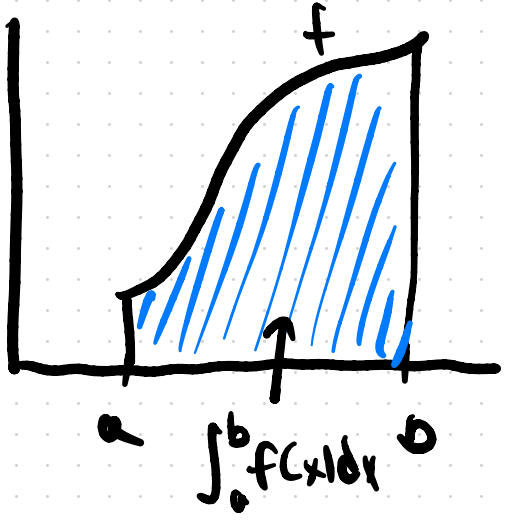
$F(y) = 2(50-y) + 5 + (8 - \frac{y}{12}) = 113 - \frac{25}{12}y$

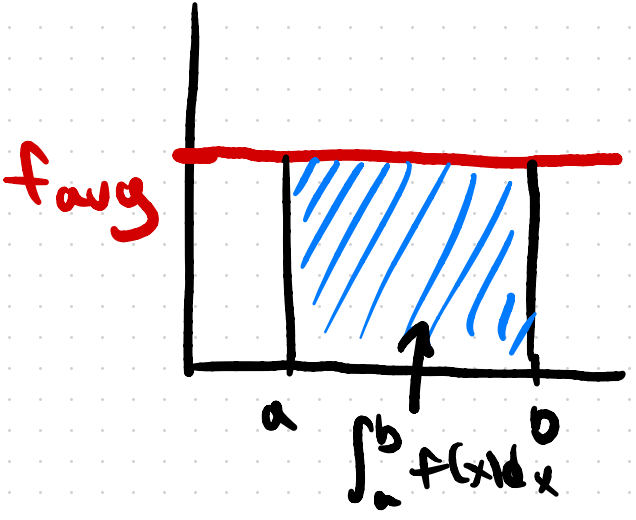
$\text{work} = \int_0^{50} F(y)\,dy = \int_0^{50} (113 - \frac{25}{12}y)\,dy$

Average value of a function (6.5)

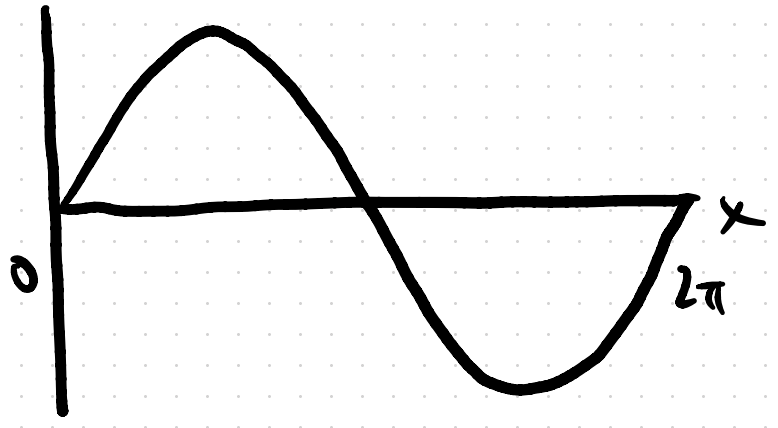
Let f be a function defined on $[a,b]$. The **average value** of f on $[a,b]$ is

$$f_{avg} = \frac{1}{b-a}\int_a^b f(x)\,dx$$





What is the average value of $\sin x$ on $[0, 2\pi]$?

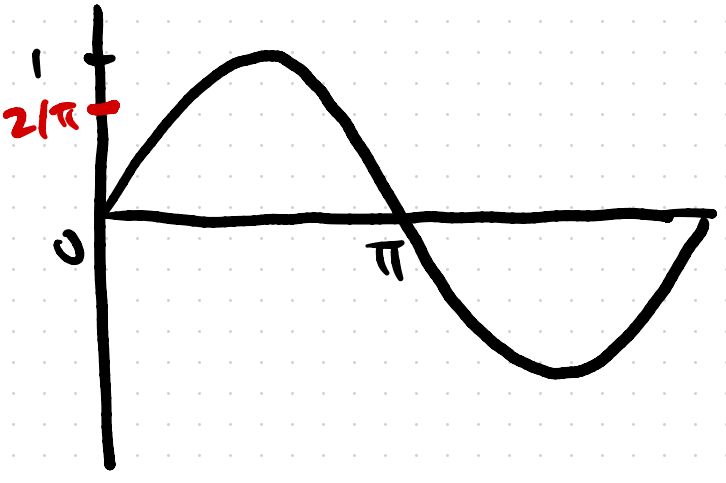


$$\frac{1}{2\pi}\int_0^{2\pi} \sin x\,dx$$

$$-\frac{1}{2\pi}\cos x\Big|_0^{2\pi}$$

$$-\frac{1}{2\pi}(1-1) = 0$$

What is the average value of $\sin x$ on $[0, \pi]$?



$$\frac{1}{\pi}\int_0^{\pi} \sin x \, dx$$

$$-\frac{1}{\pi}\cos x \Big|_0^{\pi}$$

$$-\frac{1}{\pi}(-1-1)$$

$$\frac{2}{\pi}$$

The average velocity of a car from $t=0$ s to $t=10$ s is 80 ft/s. What is the car's displacement in this time interval?

$$s(10) - s(0) = \int_0^{10} v(t)\,dt$$

$$= 10 \cdot \frac{1}{10}\int_0^{10} v(t)\,dt$$

by definition, this is the average of $v(t)$ on $[0,10]$

$$= 10 \cdot 80$$

The average velocity of a car from $t=0$ s to $t=10$ s is 80 ft/s. Is it true that at some point, the car is moving at exactly 80 ft/s?

Yes: At some point it should be moving at most 80 ft/s, and at some point it should be moving at least 80 ft/s. Since the velocity varies continuously, some time between these two times it should be moving exactly 80 ft/s.

Mean Value Theorem for integrals: If f is continuous on $[a,b]$, then there exists a number c is $[a,b]$ such that

$$f(c) = f_{avg} = \frac{1}{b-a}\int_a^b f(x)dx$$